Power Systems Agent Benchmark: Bringing Executable Evaluation to AI Agents on Electric Power Engineering Tasks

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Abstract

Executable evaluation — having an agent act and then checking the consequences of that action with a program rather than grading its prose — has become a standard way to assess tool-using AI agents in software and related settings. Electric power engineering, by contrast, has barely begun to adopt agentic AI at all: language-model use in the domain is still dominated by retrieval and text question answering, agents that act on a power system exist only as a handful of academic prototypes, and evaluation has correspondingly concerned the text a model produces rather than the consequences of an action. We bring executable evaluation to power-engineering agents. In the Power Systems Agent Benchmark an agent receives a structured task and returns a structured solution; a deterministic evaluator then recomputes the relevant engineering quantities, tests the operational constraints, and returns a feasibility flag, a normalized score, and an explicit list of violations. The benchmark spans 41 task families across eight areas of power engineering — power flow and network analysis, short circuit and protection, stability and grid-code compliance, distributed resources and storage, microgrid dispatch, reliability and restoration, power quality and standards, and forecasting — each grounded in a citable source and an engineering standard. Beyond the public set, held-out cases are not distributed as data but synthesized on demand by a per-family generator from a private seed, yielding a contamination-resistant evaluation whose construction is open to inspection while its instances remain private. Because the launch protocol is agent-agnostic, the benchmark serves as a common platform for comparing language models and agent harnesses and for studying agent behavior on physically grounded tasks; as a reference evaluation we assess three command-line agents; on this compact tier the strongest already score close to the maximum while a smaller open model trails, and public and held-out performance agree. The evaluators are compact deterministic surrogates, and the task contract is designed so that their internals can later be upgraded to simulator-backed checks without changing how tasks are posed or solved.

1. Introduction

Large language models have, over a short period, moved from systems that produce text to systems that take actions. Given access to tools, a model can read structured data, call an external program, inspect the result, and decide what to do next; wrapped in such a loop, it becomes what is now commonly called an agent. This shift has been most visible in software, where agents resolve issues in real code repositories, and in business operations, where they handle customer workflows, query databases, and execute multi-step procedures through tool and API calls. The appeal is the same across these settings: an agent can carry out work that previously required a person to translate intent into a sequence of tool invocations.

It is natural to ask whether the same capability transfers to specialized engineering domains, where the work is governed not by conventions but by physical law. Electric power systems are a demanding instance. Engineering decisions there — estimating a fault current, choosing a protective relay setting, scheduling generation under security constraints, screening a network for contingencies — rest on circuit theory, on stability and protection principles, and on a dense body of standards. Whether language-model agents can perform such work is an open research question, and an active one: recent surveys document a rapid influx of agent and LLM methods into power-system analysis [llmpowersurvey2025, agenticpowersurvey2025], and

early prototype systems have begun to appear [gridmind2025, xgridagent2025]. This activity is, for now, almost entirely experimental rather than deployed, which makes the question of how to measure progress a pressing one in its own right.

Measuring it is harder than it first appears, because correctness in power engineering is a physical property, not a textual one. A proposed relay setting can be eloquently justified and fail to be selective; a dispatch can be internally consistent and exceed a generator's limits; a restoration plan can read well and de-energize the wrong feeder. Errors of this kind are invisible to a human skimming the explanation and to a language model asked to judge it, because they are not errors of expression — they are violations of network physics and operating limits, and they surface only when the proposed solution is recomputed against the case. The two evaluation methods most common in agent research are, for precisely this reason, ill-suited here: question answering against a fixed reference answer cannot accommodate the many admissible solutions an engineering task allows, and grading a free-form report with a second model [zheng2023judging] rewards a fluent rationale even when the underlying decision is unsound — a tendency reinforced by the documented biases of such judges [wang2023fairevaluators].

In software and tool-use settings this difficulty has been met by executable evaluation: an agent's output is not scored as prose but applied and checked by a program. SWE-bench [jimenez2024swebench] runs a repository's own test suite against the agent's patch; τ-bench [yao2024taubench] compares the final state of a database against an annotated goal; terminal-bench [terminalbench2025] inspects the effect of an agent's commands on a live shell. The principle these share — judge the outcome of an action, not its description — is exactly what power-engineering evaluation needs, and it has not been applied to the domain at any breadth.

This paper does so. We introduce the Power Systems Agent Benchmark, in which each task supplies an agent with structured inputs and a required output schema; the agent returns a structured solution; and a deterministic evaluator recomputes the relevant engineering quantities, tests the hard constraints, and returns a feasibility flag, a normalized score in the unit interval, and an explicit list of violations. The agent's natural-language explanation is recorded but excluded from scoring. The public portion comprises 41 task families across eight areas of power engineering, each grounded in a citable source and an engineering standard. The held-out portion is not distributed as data: a generator produces held-out cases on demand from a private seed, so that the construction of the held-out set is open to inspection while its instances remain unknown to those being evaluated.

This benchmark is our primary contribution, and it is best understood as three things at once. It is, first, a broad executable test of power-engineering competence: 41 task families spanning eight areas of the field, each posed through a uniform contract and checked by a deterministic evaluator that recomputes the underlying physics, so that an agent is measured by what its decisions do rather than by how it describes them. It is, second, a contamination-resistant evaluation: rather than ship a fixed held-out set that ages and leaks, we provide a per-family generator that synthesizes physically consistent, solvability-checked cases from a private seed, keeping the held-out construction auditable while its instances stay private. And it is, third, an agent-agnostic platform: because participating reduces to reading a task file and writing a solution file, the same benchmark can compare different language models and different agent harnesses under one protocol, and can serve as a substrate for studying agent behavior on physically grounded tasks. We accompany the benchmark with a reference evaluation of three command-line agents and, in the course of running it, report a methodological observation — that a failure shared by all agents is a reliable signal of a defective task or evaluator — useful enough to recommend in its own right (Section 8.2). We frame this as a v0: a broad, contamination-resistant, agent-agnostic executable prototype across 41 compact task families — not a claim of priority over earlier domain efforts (Section 2), nor a measure of industrial-scale competence (Section 9).

2. Related Work

Our work draws on three lines of research, and is most easily located by where it follows each and where it departs.

The first is executable evaluation of agents. SWE-bench [jimenez2024swebench] and the earlier pass@k construction of Codex/HumanEval [chen2021codex] established that an agent's output can be judged by executing it; τ-bench [yao2024taubench] generalized this to tool-and-user interaction by comparing end states and proposed pass^k as a reliability measure; terminal-bench [terminalbench2025] and MLE-bench [chan2024mlebench] extend the idea to terminal work and machine-learning engineering, while broad suites such as AgentBench [liu2024agentbench], GAIA [mialon2023gaia], and WebArena [zhou2024webarena] probe general tool use. We adopt this paradigm; our point of difference is coverage, since none of these targets power-system engineering and its physical constraints.

The second line concerns *how* to score, and it is the reason we recompute rather than judge. What ultimately matters is the effect of an agent's actions, not the fluency of its account of them, and for verifying effects the strongest available tool is a deterministic check. This is not a new idea — it is the basis of software testing, and of competitive programming, where a submission is accepted only if it passes a fixed suite of tests — and wherever a task admits such a check, as every task in this benchmark does, it should be preferred: it is exact, reproducible, and immune to the biases that attend a language-model judge, the biases of position, length, and self-preference [zheng2023judging, wang2023fairevaluators] that are most damaging exactly when a wrong answer is fluent. Model-based judging retains a role where no deterministic check is feasible, and recent work extends it from single responses to whole agent trajectories under the heading of *agent-as-a-judge* [zhuge2024agentjudge, aisjudgeais2025]. We expect that as the benchmark grows to include scenarios too open-ended for a closed-form evaluator, judging of this kind — applied to actions and trajectories, not to prose — will become appropriate, and we leave that path open; for the tasks presented here, deterministic recomputation suffices, and the agent's rationale is kept only as a diagnostic.

The third line is domain-specific. Surveys [llmpowersurvey2025, agenticpowersurvey2025] record a rapid growth of LLM and agent methods in power systems with no settled means of evaluation. The closest effort to ours is PowerAgentBench [poweragentbench], part of the actively developed PowerAgent ecosystem at Harvard SEAS [zhang2025poweragent]. We adopt the same core design it uses — a public/held-out split with an evaluator that recomputes physical validity — which we do not claim as novel: PowerAgentBench is a prior instance of it in this domain, and contamination-resistant held-out evaluation is by now a broadly established idea [contaminationsurvey2025, white2024livebench]. ElecBench [zhou2025elecbench] scores dispatch scenarios largely through text and judges, and PFBench [pfbench2025] applies programmatic checking but only to power flow. The prototype agents now appearing — GridMind [gridmind2025], X-GridAgent [xgridagent2025], Grid-Mind CIA [gridmindcia2026], PFAgent [pfagent2026] — are the intended subjects of a benchmark of this kind. What distinguishes our work is not the split but its emphasis: breadth across many task families, held-out cases generated from a seed rather than stored, and an explicit quality-control procedure.

Two further bodies of work inform specific choices. Reinforcement-learning grid environments, principally Grid2Op/L2RPN [marot2020l2rpn, marot2021retrospective], provide a mature setting for sequential topology control and are built on established power-system simulators — pandapower [thurner2018pandapower], MATPOWER [zimmerman2011matpower], OpenDSS [dugan2011opendss], PowerModels/PGLib [coffrin2018powermodels, babaeinejadsarookolaee2019pglib], ANDES [cui2021andes], PyPSA [brown2018pypsa] — refined over many years to compute power-system physics. We address the different problem of single-shot engineering decisions across many subtasks under an agent-agnostic contract, and our current evaluators do not use these simulators at all: each is a compact, self-contained

deterministic check. We name them only as the natural backend for higher-fidelity future versions of the evaluators, not as a present dependency. Finally, the literature on benchmark integrity informs the held-out split: because training-data contamination inflates results [contaminationsurvey2025, statictodynamic2025], dynamic and refreshed benchmarks have become the standard response [kiela2021dynabench, white2024livebench, jain2024livecodebench], and a seed-driven generator is a prevention mechanism of the same family. The runner follows the prescriptions of [kapoor2024agentsmatter] on cost, reproducibility, and the separation of failure classes, and respects the limits of best-of-N characterized in [brown2024monkeys].

3. Design Requirements

The system is shaped by a small set of requirements, stated here at the outset because most of what follows is a consequence of them; we note for each where it is realized.

1. **Executable, physical verification.** A solution must be checked by running code rather than by matching strings, and that code must recompute the relevant physical quantities and test the operating constraints, not compare against stored text. The remaining requirements largely follow from this one (the contract is given in Section 4.1, the evaluators in Section 6).
2. **A leak-proof public/held-out split.** Public cases must support development and held-out cases a fair comparison; crucially, held-out answers must be excluded by construction rather than by policy, so that no amount of scrutiny of the released material reveals them (Section 4.4).
3. **Agent agnosticism.** Since the point is to compare models and harnesses on equal terms, launching an agent must not depend on any particular one — participation reduces to reading a task file and writing a solution file (Section 4.2).
4. **Provenance and machine-readable results.** Every task must declare where it comes from, with a confidence level (Section 5.1), and every result must be machine-readable — a feasibility flag, a score, and an explicit list of violations (Sections 4.1 and 6) — so that runs can be aggregated and audited without anyone reading prose.
5. **Room to grow.** The benchmark must be able to become more demanding without breaking compatibility, so that replacing a closed-form check with a simulator-backed one changes an evaluator's internals while leaving the task and solution formats untouched (Section 4.3).
6. **Specification-quality control.** Finally, because a benchmark's own tasks and evaluators can themselves be wrong, it must provide a means of telling a defect in its specifications apart from a genuine difference in agent capability (demonstrated in Section 8.2).

4. Architecture

4.1 Artifact contract

The benchmark is organized around a single linear contract (Figure 1): a task description is given to an agent, the agent produces a structured solution, an evaluator consumes the solution, and an evaluation result is emitted. A JSON schema fixes each of the three artifacts. A task carries its identity and family, its public inputs, the schema its solution must follow, the hard constraints that will be checked, the metrics, the provenance, and a bilingual presentation block. A solution carries the agent's structured answer and an optional rationale. An evaluation result carries the feasibility flag, the score, the list of violations, and any supporting evidence. Because the formats are fixed independently of how a task is solved or checked, an agent and an evaluator need share nothing beyond these schemas.

A short-circuit task, abbreviated, illustrates the form:

```
{
```



Figure 1: Figure 1: The benchmark contract. A task is posed to any command-line agent, which writes a structured solution; a deterministic evaluator recomputes the result and emits a machine-readable verdict.

```
  "family": "three_phase_short_circuit",
  "public_inputs": {"short_circuit_case": {
    "base_mva": 100.0, "voltage_kv": 10.5,
    "voltage_factor_cmax": 1.1, "voltage_factor_cmin": 1.0,
    "positive_sequence_impedances_pu": [ /* ... */ ],
    "fault_resistance_pu": 0.005}},
  "submission_schema": {
    "ik_max_ka": "max initial symmetrical current (kA)",
    "ik_min_ka": "min current with fault resistance (kA)", "...": "..."},
  "hard_constraints": [
    "maximum current uses c_max over a bolted fault",
    "values match the independent positive-sequence calculation"],
  "metrics": [{"name": "feasible", "direction": "boolean", "...": "..."},
              {"name": "score", "direction": "maximize", "...": "..."}],
  "source_provenance": [{"title": "IEC 60909 short-circuit calculation",
                         "confidence": "high", "...": "..."}]
}
```

4.2 Agent-agnostic execution and a trustworthy runner

For a comparison across agents to be meaningful, the *runner* — the benchmark's own apparatus that launches each agent (itself a harness of model, loop, and tools) on each task and scores what it returns — must satisfy three properties beyond merely invoking the agent. The first is isolation: an agent must not be able to read the evaluator or any stored answer. Each task is therefore packaged with its answer key and evaluator internals removed, and is executed in a sandbox that has no path back to the benchmark's own code. This matters because an agent with autonomous filesystem access can, in principle, reach an evaluator that is within reach and grade itself against it; a written instruction not to do so is no safeguard, since nothing compels the agent to honor it. Isolation must therefore be structural rather than instructional.

The second property is robustness to malformed output. An agent may return a solution that is syntactically valid but meaningless — null values, text where a number is required, the wrong structure entirely — and the evaluator must treat this as an ordinary failed solution, returning infeasible with a list of the offending fields, rather than raising an exception that halts the entire run. Without this property a single malformed solution can abort a multi-task evaluation partway and leave an incomplete, incomparable record.

The third property is a principled separation of failure classes. A run can fail for reasons that have nothing to do with the agent's competence — a transient provider error, a dropped connection, an exhausted usage quota — and conflating these with genuine failures contaminates the measurement. The runner classifies each failure and retries only those attributable to infrastructure, never those in which the agent produced a poor or absent solution within a healthy environment; retrying the latter would silently turn the benchmark into a best-of-N search and inflate every score. Each result records how many attempts it took and why

earlier attempts failed, and a run reports its metrics both with and without infrastructure retries, so that provider reliability is never mistaken for agent capability.

4.3 Evaluator maturity ladder

The evaluators are allowed to mature in place along a ladder: from a closed-form or graph-based surrogate, through a deterministic engineering model, to a check backed by a full power-system simulator, and ultimately to a stochastic or large-scale scenario suite. The evaluators presented here occupy the first two rungs, and within those rungs they are of two kinds. About half compute an engineering quantity in closed form from a standard or textbook method — short-circuit and symmetrical-component fault currents, an admittance matrix, a transformer thermal model, harmonic and voltage-compliance limits — and are exact for the compact problem as posed. The rest score a decision over a deliberately simplified model — a linear voltage-sensitivity stand-in for an AC power flow, or an optimization over a compact dispatch or restoration model — and are approximate by construction; the boundary is not always sharp, since some solve a linearized model exactly. None yet invokes a full power-system simulator, which is the next rung. The contract of Section 4.1 is what makes the ladder usable: because the task and solution formats are fixed independently of how a check is computed, the internal step of replacing, say, a linearized power-flow surrogate with a full AC solution upgrades an evaluator without altering anything an agent sees. The ladder describes how a check is computed, but the contract is symmetric in the tools it admits: nothing stops an agent, on its side, from invoking the same kind of software to produce a solution — a possibility today's tasks do not yet exercise but later ones will (Section 10).

4.4 The held-out split as a generator

The benchmark does not store its held-out cases at all. In place of held-out data files, it provides a generator: a collection of deterministic, physics-aware perturbations, one per family. A held-out case is produced from a public case by redrawing its numerical content — network parameters, weather series, load profiles, a fault location together with the indicator readings and customer complaints it implies — under a private random seed, and a draw is accepted only if the family's reference solver still solves it, the evaluator confirms the solution feasible, and no small perturbation of that solution scores higher — keeping the reference solution the optimum against which the case is scored (Section 6) — with a deterministic redraw otherwise; Section 8.3 confirms empirically that this yields solvable, diverse held-out instances at scale. Whoever operates a leaderboard keeps the seed private; anyone else can read exactly how held-out cases are constructed without learning the particular instances of any round. This makes the held-out set, normally the most leak-prone component of a benchmark, auditable in mechanism and private in content at the same time. The same discipline extends to the records of evaluation runs: an agent's solution to a held-out case, and the evaluator's per-case output, each disclose that case's answer, so only aggregate run summaries are published — never the per-case held-out submissions, results, or logs.

5. Task Set

The public benchmark comprises 41 task families, grouped into eight domain areas:

1. **Network analysis and operating constraints** — bus-admittance-matrix construction, corrective reactive-power dispatch for voltage limits, contingency screening [zhou2019contingency], post-contingency corrective switching, dynamic line rating [ieee738].
2. **Short circuit and protection** — three-phase and single-line-to-ground fault calculation [iec60909, anderson1995faulted], breaker and relay setting from fault levels, distance-protection zone setting and overcurrent coordination [horowitz2014relaying], cable sizing [iec60364552].

3. **Stability, grid code, and inverter-based resources** — critical clearing time by the equal-area criterion [kundur1994stability], transient-stability classification, fault-ride-through compliance [ieee2800, entsoerfg], current-limited inverter fault response, minimum synchronous-generation share under inertia and short-circuit constraints.
4. **Distributed resources, PV, EV, and storage** — Volt-VAR control [turitsyn2010voltvar], vehicle-to-grid scheduling and voltage support, battery ancillary-service response, soft-cost levelized-cost calculation.
5. **Microgrids and dispatch** — economic and rolling-horizon dispatch, islanded active/reactive dispatch, robust dispatch under uncertainty, hydro-thermal unit commitment, single-contingency-secure generation commitment [scuccts2020].
6. **Reliability and restoration** — fault isolation and service restoration [restoration2020], faulted-section localization, fault-indicator placement, reliability-index improvement [ieee1366], operator switching and shedding.
7. **Power quality, standards, assets, and cybersecurity** — voltage-compliance checking to a supply standard [en50160], power-quality event classification, harmonic-limit compliance [ieee519], transformer thermal loading and insulation ageing [iec600767], bad-data identification in state estimation [abur2004se], secured-measurement placement [liu2011fdi], lead-station handover in a multi-agent frequency scheme.
8. **Forecasting under uncertainty** — wind power forecasting, prediction-interval estimation.

Each task was checked for physical correctness against its cited source before inclusion. Every family carries a documented source and confidence level (Section 5.1), and the appendix lists the full catalog with identifiers.

5.1 Provenance and confidence

Because the task set is drawn from the engineering literature rather than authored from scratch, each family records where its formulation comes from and how firmly it is tied to that origin. Every task carries a provenance entry — a source title, a locator (a digital object identifier or URL for an article, a clause for a standard, or a full citation for a textbook), and a short note on how the source maps onto the executable case — together with a stated confidence level. The confidence level is a deliberate, declared judgment rather than a measured quantity, and we assign it by a fixed rubric so that it is reproducible:

- **high** — the task reproduces a method defined in a published standard or a canonical textbook, cited specifically, and the executable case follows that method directly (for example, the short-circuit [iec60909], transformer-loading [iec600767], and harmonic-compliance [ieee519] tasks follow named international standards, and the critical-clearing-time [kundur1994stability] and bad-data [abur2004se] tasks follow standard textbook procedures);
- **medium** — the task is tied to an identified primary source, or to a universally taught engineering method written out as its governing equations, and the case is a faithful compact instance of it;
- **low** — no single primary document is pinned: the method is standard but the task is an adaptation in its spirit rather than a reproduction of a specific source.

Under this rubric, of the 41 families 11 are rated high, 28 medium, and 2 low; the two low cases (uncertainty-aware dispatch and minimum synchronous share) keep that rating because no single source we could verify fully grounds their exact formulation, and we prefer to say so than to inflate the label. We regard the scheme as a floor on transparency rather than a guarantee of fidelity: it lets a reader trace any task to its origin and see at a glance how tightly it is bound there, and it makes explicit which tasks are faithful renderings of a standard and which are looser adaptations of common practice.

6. Evaluators and Metrics

Every evaluation returns the same four things: a feasibility flag, a score in the unit interval, a list of violations stated in words, and task-specific evidence. The two numbers play distinct roles. Each task declares a set of *hard constraints* — the physical and operational requirements a solution must satisfy to be admissible at all, such as respecting a generator's limits, holding every voltage within its band, or keeping the network radial — and a solution is *feasible* only if it violates none of them. The score then grades quality among feasible solutions; an infeasible solution scores zero however close it may have looked.

How the score is computed depends on the task, but every score is reported *relative to the reference optimum*: it is scaled so that an optimal solution earns exactly one, and a feasible but worse solution earns a fraction below it. This matters because many tasks have no natural unit ceiling on their raw objective — a reactive effort in kvar, a switching cost, an amount of restored load — so without this scaling even a perfect answer would score below one. The reference solver for each family computes that optimum and is optimal by construction; an agent's solution, being one feasible solution among many, cannot exceed it, and should a submission nonetheless score higher the reference was not optimal on that case and the reported score is capped at one. An exact computation is scored against numeric tolerances — a 0.1% relative band with a small absolute floor, so that an answer correct to about four significant figures passes while a genuinely wrong one does not; a control or optimization task scores the gap between the submitted objective and the optimum, so that an alternative optimum of equal cost receives full credit and a feasible but costlier solution receives partial credit; a classification or localization task is scored against exact sets with enumerated misses; and a standards-compliance task as pass/fail with diagnostic fields.

A microgrid dispatch task makes the distinction concrete: its hard constraints are the power balance, the generator and battery limits, and the state-of-charge bounds; any schedule meeting all of them is feasible, and among feasible schedules the least-cost schedule scores exactly one while a costlier feasible dispatch scores below one in proportion to its cost gap.

At the level of a whole run we report two aggregates that need not agree: the *feasible rate*, the fraction of tasks whose solution satisfied every hard constraint, and the *perfect rate*, the fraction scoring exactly one (feasible and, where graded, optimal). The two diverge whenever an agent produces admissible but suboptimal solutions, and because they can rank agents differently we report both alongside the mean score.

7. Reproducibility

Every number in this paper can be regenerated from the repository. A single command runs the unit tests over the evaluators, the reference solvers, the runner, and the generator; validates every artifact; evaluates each of the 41 public sample solutions individually; and runs the reference solvers across the public set, which return all 41 cases feasible. Because each task's score is measured relative to its own reference optimum (Section 6), a reference solution scores one by construction, so the check confirms not that the tasks are easy but that every task has a contract-valid, physically consistent reference solution. One limit of such a check is worth stating plainly: a reference solver and an evaluator built on the same kernels can share the same mistake and still agree, so their agreement confirms the contract but cannot, by itself, certify the evaluator's underlying physics. As a partial, independent check on that physics, a set of evaluators is additionally cross-checked against golden cases whose answers are derived by hand from the governing formulas rather than from our own baselines — at present three: three-phase short circuit (IEC 60909), single-line-to-ground fault by symmetrical components, and admittance-matrix construction. Continuous integration runs the full check on every change.

8. Experiments

We evaluated three command-line agents on an identical task input — the same `task.json` and the same instruction file for every agent on every task — under a fixed per-task timeout and the infrastructure-retry policy of Section 4.2, with no adaptation of the agents to the benchmark. Each agent was invoked through its own command-line interface and ran under whatever internal system prompt and scaffolding it ships with; we standardize what the task asks, not how an agent organizes itself to answer it, since that scaffolding is part of what an agent is and what the benchmark compares. The three are Codex (codex-cli 0.139.0) running gpt-5.5 at medium reasoning effort, Cursor's command-line agent (cursor-agent 2026.06.12) running composer-2.5, and the open-source OpenCode agent (opencode 1.15.13) running deepseek-v4-flash-free, the free tier of that model through OpenCode's provider; we refer to them below as Codex, Cursor, and OpenCode. Both the models and these agent CLIs are updated frequently, so the round, run in June 2026, should be read as a snapshot of these specific versions rather than a standing ranking; re-running the same protocol with later releases may give different results.

8.1 Public versus held-out evaluation

Each agent attempted the public case of every family and one freshly generated held-out case per family from a new private seed, every solution scored by the same deterministic, optimum-normalized evaluators.

agent	public feasible	public perfect	public mean	held-out feasible	held-out perfect	held-out mean
Codex	40/41	38/41	0.973	41/41	40/41	0.995
Cursor	40/41	39/41	0.973	41/41	41/41	1.000
OpenCode	39/41	39/41	0.951	40/41	39/41	0.973

Two things stand out, and the first is sobering for a benchmark: the strongest agents score so close to the maximum that it has little room left to tell them apart. Codex and Cursor are nearly indistinguishable on the public split and essentially solve the held-out one, and OpenCode trails only modestly. We therefore do not present this round as a sharp ranking of frontier agents; on this compact tier the benchmark has limited power to separate the best of them, a point we take up in Section 9.

Second, the two splits agree — every agent scores at least as high on the freshly generated held-out cases as on the public ones. The signature one would expect from contamination, public scores standing above held-out, is absent; if anything the held-out draws were marginally easier. We make no claim of an intrinsic difficulty gap between the splits, and the case for the generator is prospective: a published set ages as models train on it, and the generator is what keeps an uncontaminated draw available.

What discrimination the benchmark does provide comes at the lower end of the capability range and through shared blind spots, not through margins among the leaders: OpenCode fails a microgrid-dispatch task outright where the others succeed, and one storage-scheduling task is missed by every agent in the same way — a genuine capability gap, confirmed not to be a specification defect (Section 8.2).

8.2 Distinguishing capability from specification

Reading the failures across agents rather than per agent proved the most useful analysis available, because the failures separate into two kinds. When agents fail a task in different ways — one breaching a voltage limit where the others hold it, one returning a wrong ride-through verdict, one inverting a sign — the disagreement is signal, and it is what a leaderboard should reward. But when every agent fails a task in the same way, the

task itself, rather than the agents, is the first thing to suspect. Treating that pattern as a diagnostic during construction surfaced a series of specification defects — mismatches between a field's description and the value the evaluator expected, unstated conventions, and over-strict comparisons that rejected admissible answers — each confirmed by correcting the specification and watching the agents recover. The pattern is a suspicion, not a verdict: a unanimous failure is sometimes a shared blind spot rather than a defect, so it should be confirmed by inspection. We propose this rule — treat a unanimous failure as a likely defect in the specification or the evaluator, then confirm or dismiss it — as a routine quality-control step for executable benchmarks.

The most consequential instance pointed not at a task but at an evaluator, and is worth stating in full, as it is the clearest evidence we have for the value of the method. On one held-out reliability case concerning faulted-section localization, all three agents identified the same faulted section, a conclusion consistent both with the fault-current-indicator readings and with the pattern of customer complaints; the evaluator demanded a different section that in fact contradicted the case's own indicator readings. The agents were correct and the evaluator was wrong. The cause was an off-by-one error in the evaluator's indicator model: it had placed the faulted section's downstream node on the fault-current path, so that an indicator at that node was predicted to register a fault current it would not in fact see, whereas an indicator registers fault current only when the fault lies strictly downstream of it. What makes this more than a corrected bug is that the benchmark's own safeguards could not have caught it. The reference solver used for self-checking is derived from the same function as the evaluator, so solver and evaluator agreed with each other and the public sample passed; the error was invisible from within and became visible only because three independent agents brought the correct physics to bear. An executable benchmark, in other words, can harbor a latent domain error in the very code that defines correctness, and a matrix of agents is a means of finding it — a caution we would extend to any benchmark whose reference answers are derived from its evaluator.

8.3 Held-out generator validation

The held-out mechanism of Section 4.4 is only useful if the generator reliably produces solvable, varied instances rather than near-duplicates of the public case. We tested this directly by drawing eight held-out variants of every family from a fixed validation seed and scoring each with the reference solver. All 328 variants (41 families × 8) were produced without exhausting the redraw budget and were feasible, so the acceptance rule — keep a draw only if the reference solver solves it and no small perturbation of that solution scores higher — holds at scale. Every variant differed from its public case, and 40 of the 41 families yielded eight distinct instances; the exception, faulted-section localization, draws a fault location from a small discrete set and so produces fewer distinct instances (four of eight) — an inherent bound for categorical families rather than a generator fault. Across the continuous content the perturbed inputs varied with a mean coefficient of variation near 0.2, so the variants are genuinely different problems. This validates the generator itself: it is a property of the held-out construction, not a measure of agent performance, for which repeated agent runs over many variants — beyond the single configuration reported above — would be required.

8.4 A wider spectrum of accessible models

To check that the benchmark separates more than the three reference agents, we ran four further, widely accessible models on the public split through a single model-agnostic harness, so that only the model varies. On the same optimum-normalized evaluators their public means span a wide range below the frontier: Kimi K2.5 (a ~1-trillion-parameter open-weight mixture-of-experts, 32B active) at 0.93, GLM-4.6 (357B, 32B active, open) at 0.78, Claude Haiku 4.5 (proprietary; size undisclosed) at 0.71, and Qwen3-Coder-Next (80B, 3B active, open) at 0.66, against the reference agents at 0.95–0.97. The benchmark therefore discriminates clearly across the realistic spectrum of models a practitioner might actually deploy, not only at the frontier;

the ordering also tracks something other than raw size, as the two 32B-active models differ by 0.15. Three caveats keep this honest: all four share one harness, so this varies the model and not the scaffolding; each is a single run; and some failures are infrastructure rather than capability — two models occasionally wrote no solution at all (counted as infeasible), which depresses their scores below their true competence. The genuine failures are of the kind the benchmark is meant to catch: a fault current reported a thousand times too small, dispatch schedules that breach state-of-charge limits, a missed islanding flag, an inverted ride-through verdict.

9. Limitations

Several limitations bound what these results support. The public cases are compact — networks of a few buses and short time series — so they test whether an agent commands a method, not whether it scales to industrial data volumes. The evaluators are deterministic surrogates, and no claim of industrial AC accuracy is warranted until simulator-backed checkers are in place. With one case per family per split, a ranking over 41 points is statistically fragile, though the generator makes enlarging the sample inexpensive. The tasks are also self-contained: an agent computes each answer directly and is never required to operate professional power-system software, so the benchmark does not yet measure mastery of the simulators and tools that much real engineering work depends on. A reference solution scores one by construction, which is a self-consistency check on the contract, not a lower bound on difficulty, and cannot validate the evaluator. The quality-control method of Section 8.2 — reading a unanimous agent failure as a likely defect — has a limit that grows with the benchmark's difficulty: it can tell a broken task from a merely hard one only while the agents are strong enough that genuine failures are rare, so as tasks approach the frontier and honest failures become common, a unanimous failure no longer points clearly at the specification, and the unsolved tail must then be validated by independent recomputation rather than by agent agreement. And because each score is reported relative to its task's reference optimum, the scale is only as trustworthy as that reference's optimality: we verify optimality when a held-out case is generated and treat any solution that beats the reference as a flag rather than a score above one, but we do not prove global optimality for every family, so any future continuous-objective task would need its reference established with the same care. The seed of a held-out round is spent once the round is disclosed, so a citable leaderboard requires a fresh seed on an isolated host. The sandbox isolates by relocation rather than containerization, which removes leakage through relative paths and environment but not a determined search for the evaluator by absolute path, so competitive use would want host-level containers; packaging the benchmark as an environment for a container-based evaluation harness such as Harbor [harbor2026] is a natural route to that isolation, and would additionally supply the parallel execution and rollout interfaces needed to estimate the sampling variance we currently leave unmeasured. We run one configuration per agent and so do not estimate the variance of model sampling.

A broader question is what the benchmark is worth when the strongest current agents already solve most of its public tasks. On this compact tier its power to rank the leading agents is limited, and we do not present it as a frontier-stumping leaderboard; its value rests elsewhere. It separates the realistic spectrum of agents rather than only the leaders, with OpenCode, on a small open model, trailing the frontier clearly and failing outright where the leaders succeed. It surfaces shared blind spots even when scores are high, because a task that every agent misses in the same way is a capability gap an executable check makes visible and a prose grade would not. Its contract is built to grow into the harder tiers of Section 10, where an agent must operate professional simulators rather than compute compact instances and where the headroom is large. And its resistance to contamination is prospective, the generator keeping an uncontaminated draw available as the public set ages and is trained on. The contribution is a growing, physics-grounded, contamination-resistant yardstick for a domain that had only textual assessment, not a one-round wall of difficulty.

10. Conclusion

We have brought executable evaluation, by now standard for software and tool-use agents, to electric power engineering, where assessment had remained textual. The benchmark turns power-system tasks into checks that recompute physics; on a matrix of current agents the strongest already score close to the maximum on the present compact tier while OpenCode, a smaller open model, trails, and public and held-out performance agree. Its most transferable contribution is not any single task but a method: reading a unanimous failure as a defect signal, which over the course of this work repeatedly uncovered specification defects and even a latent error in an evaluator's own physics that the benchmark could not have caught by self-consistency alone — a reminder that a reference answer derived from an evaluator cannot certify it. The lasting value of the artifact lies in the contract, the domain-grounded task set, the evaluators, the seed-driven held-out generator with its prospective resistance to contamination, the isolated runner, and the reproducible quality-control procedure; one line of work ahead is to fill the evaluators with simulator-backed implementations without disturbing the contract that holds them together.

A second axis of growth concerns the agent rather than the evaluator. The tasks presented here are self-contained: an agent reads a case and computes an answer, and while it may write and run scratch code, no task yet requires it to operate professional power-system software. Yet much of real engineering work is exactly that — driving a load-flow package, a short-circuit tool, or a dynamic simulator — and the production agents now emerging in the domain already orchestrate such tools. The contract imposes no limit here: an agent is free to invoke whatever tools its sandbox provides to produce a solution. A natural next generation of tasks would therefore measure not closed-form competence but mastery of these specialized environments, with the simulator serving as the agent's instrument rather than the evaluator's. The duality is worth naming: the same simulator can sit on either side of the contract — the agent's tool for solving, or the evaluator's backend for checking — and the isolation of Section 4.2 is exactly what must keep the two apart, so that an agent can never reach the checking instance as an oracle.

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Appendix A. Task Catalog

The 41 families are listed below by domain area, each with its primary source or governing standard and the confidence level of Section 5.1 (11 high, 28 medium, 2 low). The benchmark ships the full record for every family in English and Russian: the complete task contract — inputs, objective, hard constraints, and submission schema — together with the provenance entry (source title, locator, and the note mapping the source onto the executable case).

1. Network analysis and operating constraints

Family	Source / standard	Confidence
`ybus_construction`	Glover, Overbye & Sarma (textbook)	high
`pf_voltage_repair`	Glover, Overbye & Sarma (textbook)	high
`contingency_screening`	Graph-computing contingency screening	medium
`corrective_transmission_switching`	SCLIC with corrective switching (arXiv:2001.00597)	medium
`dynamic_thermal_rating`	IEEE 738; Safari et al., probabilistic DTR	medium

2. Short circuit and protection

Family	Source / standard	Confidence
`three_phase_short_circuit`	IEC 60909	high
`earth_fault_calculation`	Anderson, Analysis of Faulted Power Systems	medium
`breaker_relay_short_circuit`	IEC 60909; Glover, Overbye & Sarma	high

Family	Source / standard	Confidence
distance_protection_settings	Horowitz & Phadke, Power System Relaying	medium
overcurrent_protection	Horowitz & Phadke, Power System Relaying	medium
cable_ampacity_sizing	IEC 60364-5-52 / IEC 60949	high

3. Stability, grid code, and inverter-based resources

Family	Source / standard	Confidence
critical_clearing_time	Equal-area criterion (Kundur)	high
transient_stability_prediction	Chen et al., transient-stability prediction	medium
frt_compliance	IEEE 2800 / ENTSO-E RfG (FRT)	medium
ibr_short_circuit_frt	IBR modeling for short circuit / FRT	medium
min_synchronous_share	Wen et al., low-inertia frequency stability	low

4. Distributed resources, PV, EV, and storage

Family	Source / standard	Confidence
pv_volt_var	Turitsyn et al., local Volt-VAR control	medium
ev_v2g_outage_schedule	EVs for power quality & security	medium
ev_v2g_voltage_support	EVs for power quality & security	medium
bess_ancillary_response	Gonzalez-Longatt & Rueda Torres	medium
commercial_pv_lcoe_uncertainty	PV soft-cost uncertainty (NREL)	medium

5. Microgrids and dispatch

Family	Source / standard	Confidence
microgrid_economic_dispatch	España et al., microgrid dispatch	medium
rolling_microgrid_dispatch	España et al., microgrid dispatch	medium
islanded_microgrid_pq_dispatch	España et al., microgrid dispatch	medium
dispatch_uncertainty	Chung, advanced prediction for smart grids	low
hydro_thermal_storage_uc	ADMM unit commitment	medium
n1_generation_commitment	SCUC with corrective switching (arXiv:2001.00597)	medium

6. Reliability and restoration

Family	Source / standard	Confidence
`flisr_restoration`	Two-stage distribution service restoration	medium
`fault_section_localization`	Brown, faulted-circuit indicators	medium
`fci_placement`	Brown, faulted-circuit indicators	medium
`fci_saidi_caidi`	Brown, FCIs; IEEE 1366 indices	medium
`operator_breaker_load_actions`	Glover, Overbye & Sarma (textbook)	medium

7. Power quality, standards, assets, and cybersecurity

Family	Source / standard	Confidence
`en50160_voltage_compliance`	EN 50160	high
`power_quality_event_classification`	IEC 61000-4-30	high
`harmonic_ieee519_compliance`	IEEE 519	high
`transformer_thermal_loading`	IEC 60076-7	high
`fdi_state_estimation`	Abur & Exposito, state estimation	high
`protected_meter_placement`	Graphical defense vs false-data injection	medium
`mas_frequency_handover`	Multi-agent control of active distribution networks	medium

8. Forecasting under uncertainty

Family	Source / standard	Confidence
`wind_power_forecast`	Safari et al., short-term wind forecasting	medium
`wind_prediction_interval`	Khorramdel et al., fuzzy prediction intervals	medium

Appendix B. Experiment Artifacts

Run configuration. Each agent was invoked once per case through its own command-line interface under a 600-second per-case timeout and the infrastructure-retry policy of Section 4.2 — at most two attempts, retrying only provider- or transport-transient failures, never quality, contract, or timeout failures. One public case and one freshly generated held-out case were run per family, and every submission was scored by the deterministic, optimum-normalized evaluators of Section 6. The versions were Codex on gpt-5.5 at medium reasoning effort (codex-cli 0.139.0), Cursor's command-line agent on composer-2.5 (cursor-agent 2026.06.12-01-15-52-7244546), and OpenCode on deepseek-v4-flash-free (opencode 1.15.13).

Public shortfalls. Every public case not scored one is as follows. Codex missed contingency_screening (0.9) and returned an infeasible ev_v2g_outage_schedule, with pv_volt_var within 10^{-5} of the optimum but short of an exact one; Cursor missed contingency_screening (0.9) and ev_v2g_outage_schedule (infeasible);

OpenCode returned infeasible solutions for ev_v2g_outage_schedule and pv_volt_var. All remaining public cases scored one.

Distribution. The reference round ships with the benchmark as per-agent, per-split run summaries. For the public split the per-case submissions, evaluator results, and agent logs are included; for the held-out split only the aggregate summaries are published, since a per-case held-out submission or evaluator result would disclose that case's answer (Section 4.4).